You must include TWO paragraphs that you print and attach to your poster:

Paragraph 1: The Story of Horse Evolution

- Use **VIDA thinking. Turn in a VIDA chart to Ms. Broo, but DO NOT put on your poster**
- Explain:
  - Variation in teeth and body size
  - Environmental change (forests → grasslands)
  - Differential survival/reproduction
  - How populations changed over time
- Use correct scientific language (avoid “needed to,” “tried to,” etc.)

Paragraph 2: Museum Misconception Analysis + Solution

In many museum exhibits, evolution is shown incorrectly. Your job is to **identify AND fix these problems**. In your second paragraph:

1. Identify **TWO common misconceptions**, such as:
   - Evolution is linear (a straight line or “ladder”)
   - Evolution has a goal or leads to “better” organisms
   - Modern horses are the final, most advanced form
2. Explain **why each misconception is incorrect** using scientific reasoning
3. **Explain how YOUR display corrects each misconception**, using specific features of your poster:
   - Branching phylogenetic tree
   - Overlapping timelines of species
   - Multiple species existing at the same time
   - Environmental context (changing habitats)

1 POINT	3 POINTS	8 POINTS
VIDA chart is incomplete or inaccurate. Paragraphs are missing or unclear. Misconceptions are missing or incorrectly explained. No connection to how the display addresses misconceptions.	VIDA chart is mostly complete but lacks depth or clarity. Evolution paragraph is partially accurate. Misconceptions are identified but explanation is vague OR connection to display is weak or unclear.	VIDA chart is complete, accurate, and clearly explains Variation, Inheritance, Differential Survival, and Adaptation using evidence from teeth and body size. First paragraph clearly explains horse evolution using natural selection. Second paragraph identifies at least TWO common misconceptions and clearly explains why they are incorrect AND explicitly explains how the student’s museum display corrects each misconception using evidence (e.g., branching patterns, overlapping timelines, multiple coexisting species).

Example: VIDA Chart + Paragraphs (Rock Pocket Mice)

VIDA Chart Example

V – Variation
In the rock pocket mouse population, there is variation in fur color. Some mice have light-colored fur and others have dark-colored fur due to genetic mutations.

I – Inheritance
Fur color is a heritable trait passed from parents to offspring through genes.

D – Differential Survival/Reproduction
On light-colored desert rocks, light mice are better camouflaged and less likely to be eaten by predators. On dark lava flows, dark mice are better camouflaged and survive at higher rates. Mice that survive are more likely to reproduce and pass on their fur color alleles.

A – Adaptation
Over time, populations living on dark lava flows become mostly dark-colored, while populations on light rocks remain mostly light-colored. This change in trait frequency is an adaptation to the environment.

Model Example (Rock Pocket Mice – Honors-Level)

Paragraph 1 (VIDA Summary)

Rock pocket mice provide a clear example of how environmental change drives evolution through natural selection. Within the population, there is variation in fur color due to genetic mutations, and this trait is inherited from parents to offspring. When volcanic activity created dark lava flows in parts of the desert, the environment changed from light-colored sand and rocks to much darker substrates. This shift introduced a new selective pressure from visual predators, such as birds, that more easily detected mice that contrasted with their surroundings. On dark lava, light-colored mice were more visible and more likely to be eaten, while dark-colored mice were better camouflaged and more likely to survive and reproduce. Over generations, the frequency of the dark fur allele increased in these populations. In contrast, populations living on light-colored desert surfaces remained mostly light because those individuals had higher survival in that environment. This demonstrates that environmental changes alter selective pressures, leading to changes in allele frequencies in populations over time rather than individuals changing in response to need.

Paragraph 2 (Misconception Analysis)

A common misconception is that rock pocket mice changed color because they needed to survive on dark lava rocks, suggesting that evolution is goal-directed. In reality, dark fur already existed as a genetic variation within the population, and natural selection increased its frequency over time. Another misconception is that all mice in a population change at the same time, rather than populations shifting in allele frequency across generations. A museum display could correct these misconceptions by showing multiple mice with different fur colors existing at the same time and using a branching diagram to illustrate how different populations evolved under different environmental conditions. This emphasizes that evolution acts on existing variation and that populations (not individuals) evolve over time.